

PAIC simulation results

2023-12-19

Overview of the problem

The goal is to estimate the relative effect of treatment A vs B , noted Δ_t . These two treatments have been evaluated in two distinct trials, AB and AC . The target estimand is the relative effect of A vs B in the source population of trial AC , written $\Delta_{t(AC)}$.

$\Delta_{t(AB)}$ will be estimated using different estimators:

- Unadjusted:
 - Unanchored: naive model
 - Anchored: difference of relative effects A versus C and B versus C
- Adjusted
 - Adjusted using a multivariable outcome model (regression model), or univariable weighted outcome model (propensity score)
 - * Each of these models will be fitted using either Individual Patient Data (IPD) for both trials, or IPD for the trial AB and AgD for trial BC : Population-Adjusted Indirect Comparisons (PAIC). PAIC multivariable outcome model = Simulated Treatment Comparison (STC), univariable weighted outcome model = Matching-Adjusted Indirect Comparisons (MAIC).
 - * Each of these models will be either anchored or unanchored.

Summary of the estimators

Estimand	Common comparator?	Full IPD?	Estimator
Marginal	Yes	Yes	NMA with IPTW PS
Marginal	No	Yes	IPTW PS
Marginal	Yes	No	Anchored MAIC
Marginal	No	No	Unanchored MAIC
Conditional	Yes	Yes	IPD NMA
Conditional	No	Yes	Regression-based outcome model
Conditional	Yes	No	Anchored STC
Conditional	No	No	Unanchored STC

Parameters of the different scenarios

- **covariate_names**: Covariates included in the propensity score model
- **outcome_regression_model**: Outcome model used in regression-based estimators
- **model_marginal**: Is the marginal model (propensity score) adequate? Or is it missing variables (confounded), or including non confounding factors variables (imprecise)?
- **model_conditional**: Is the outcome regression model adequate? Or is it missing confoundings, or including non confounding variables?
- **any_nonnormal**: Are at least one of the PF/TEM non-normally distributed?

Table 2: Parameters of different scenarios

scenario_num	covariate_names	outcome_regression_model	model_marginal	model_conditional	any_nonnormal
1	X1	$Y_{\text{obs}} \sim X1^{*ttt}$	confounded	confounded, imprecise	FALSE
2	X2	$Y_{\text{obs}} \sim X2^{*ttt}$	confounded	confounded, imprecise	FALSE
3	X1, X2	$Y_{\text{obs}} \sim X1^{*ttt} + X2^{*ttt}$	adequate	imprecise	FALSE
4	X1	$Y_{\text{obs}} \sim X1^{*ttt}$	confounded	confounded	FALSE
5	X2	$Y_{\text{obs}} \sim X2^{*ttt}$	confounded	confounded, imprecise	FALSE
6	X1, X2	$Y_{\text{obs}} \sim X1^{*ttt} + X2^{*ttt}$	adequate	imprecise	FALSE
7	X1	$Y_{\text{obs}} \sim X1^{*ttt}$	confounded	confounded, imprecise	FALSE
8	X2	$Y_{\text{obs}} \sim X2^{*ttt}$	confounded	confounded	FALSE
9	X1, X2	$Y_{\text{obs}} \sim X1^{*ttt} + X2^{*ttt}$	adequate	imprecise	FALSE
10	X1	$Y_{\text{obs}} \sim X1^{*ttt}$	confounded	confounded	FALSE
11	X2	$Y_{\text{obs}} \sim X2^{*ttt}$	confounded	confounded	FALSE
12	X1, X2	$Y_{\text{obs}} \sim X1^{*ttt} + X2^{*ttt}$	adequate	adequate	FALSE
13	X1	$Y_{\text{obs}} \sim X1^{*ttt}$	confounded	confounded, imprecise	TRUE
14	X2	$Y_{\text{obs}} \sim X2^{*ttt}$	confounded	confounded, imprecise	TRUE
15	X1, X2	$Y_{\text{obs}} \sim X1^{*ttt} + X2^{*ttt}$	adequate	imprecise	TRUE
16	X1	$Y_{\text{obs}} \sim X1^{*ttt}$	confounded	confounded	TRUE
17	X2	$Y_{\text{obs}} \sim X2^{*ttt}$	confounded	confounded, imprecise	TRUE
18	X1, X2	$Y_{\text{obs}} \sim X1^{*ttt} + X2^{*ttt}$	adequate	imprecise	TRUE
19	X1	$Y_{\text{obs}} \sim X1^{*ttt}$	confounded	confounded, imprecise	TRUE
20	X2	$Y_{\text{obs}} \sim X2^{*ttt}$	confounded	confounded	TRUE
21	X1, X2	$Y_{\text{obs}} \sim X1^{*ttt} + X2^{*ttt}$	adequate	imprecise	TRUE
22	X1	$Y_{\text{obs}} \sim X1^{*ttt}$	confounded	confounded	TRUE
23	X2	$Y_{\text{obs}} \sim X2^{*ttt}$	confounded	confounded	TRUE
24	X1, X2	$Y_{\text{obs}} \sim X1^{*ttt} + X2^{*ttt}$	adequate	adequate	TRUE

Data Generation Model :

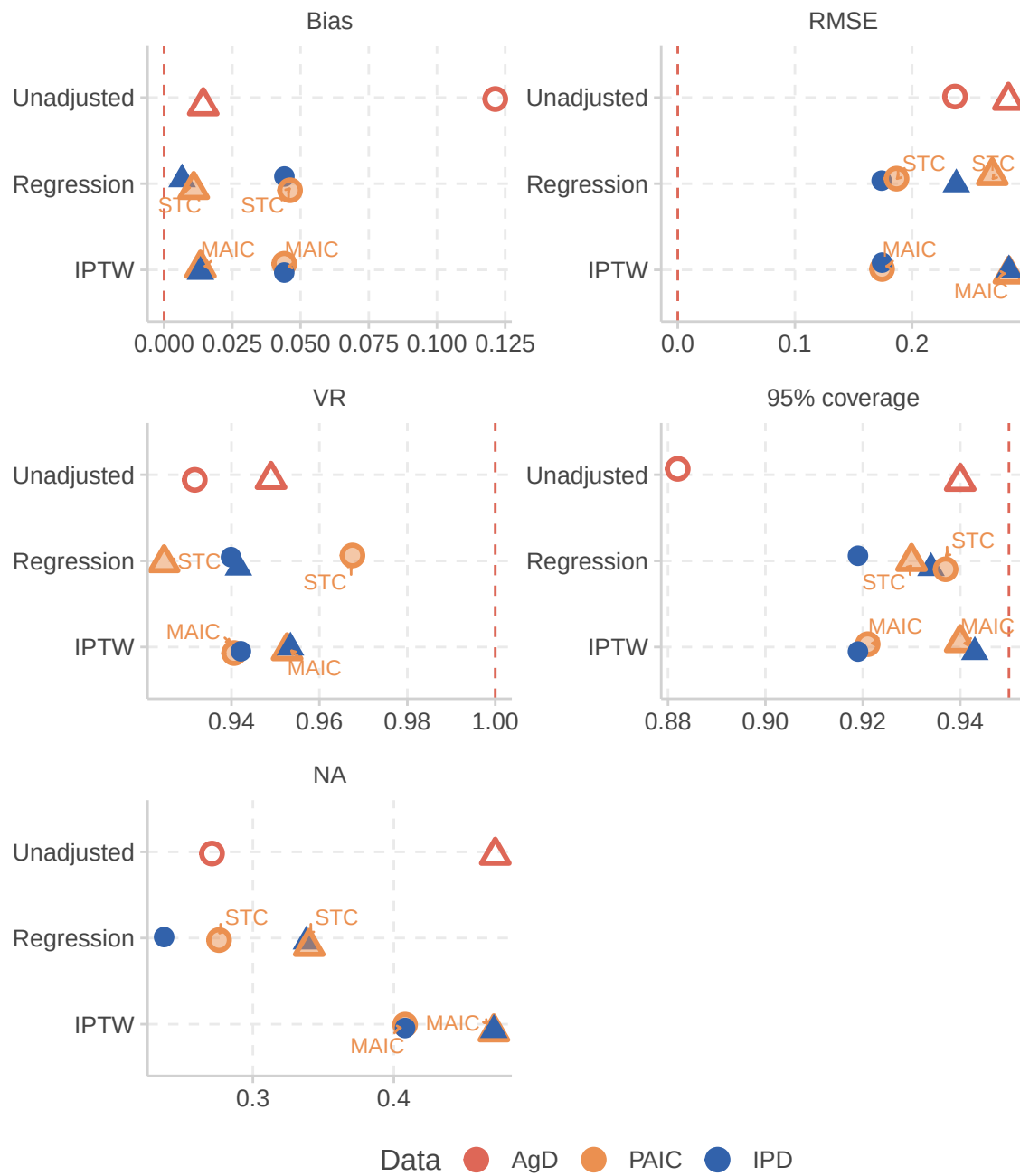
$$Y = \beta_{Y_{X1}} \times X_1 + \beta_{Y_{X2}} \times X_2 + (\beta_{Y_A} + \beta_{Y_{A,X1}} \times X_1 + \beta_{Y_{A,X2}} \times X_2) * A + \beta_{Y_B} \times B + \beta_{Y_C} \times C + \epsilon$$

With $\epsilon \sim N(0, 1)$

Performance of the estimators

Scenario number 1

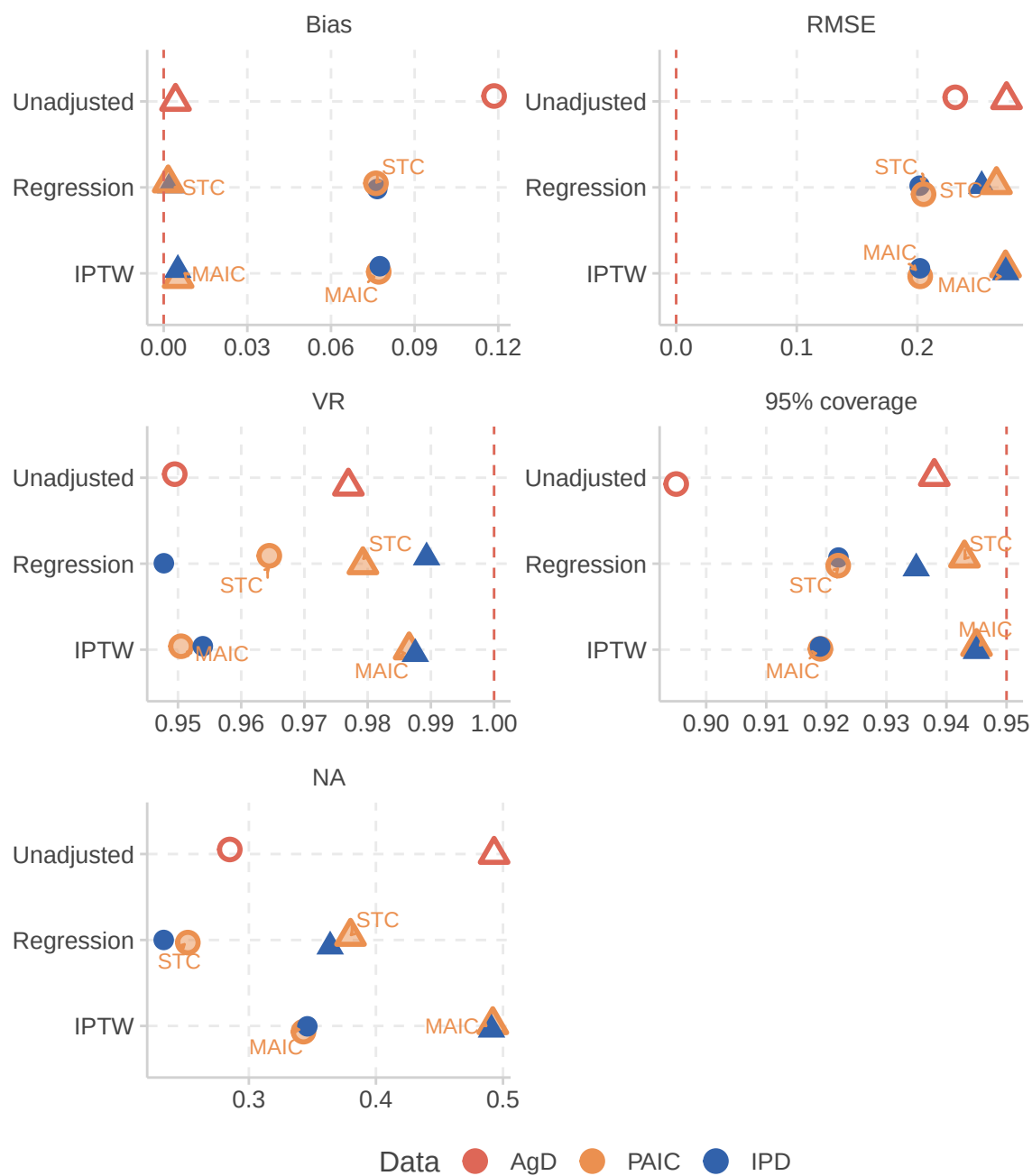
bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	0	0	rbinom(N_pop, 1, 0.5)	rnorm(N_pop, 0.5, 1)	X1	$Y_{obs} \sim X1 * ttt$



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	0.013	0.283	0.953	0.940	0.471
		Unanchored	PAIC	0.044	0.174	0.941	0.921	0.408
	ML	Anchored	IPD	0.013	0.283	0.953	0.943	0.471
		Unanchored	IPD	0.044	0.174	0.942	0.919	0.408
Regression								
	GLM	Anchored	IPD	0.007	0.238	0.942	0.934	0.338
		Unanchored	IPD	0.044	0.174	0.940	0.919	0.237
	STC	Anchored	PAIC	0.011	0.268	0.925	0.930	0.340
		Unanchored	PAIC	0.046	0.187	0.968	0.937	0.276
Unadjusted								
	Unadjusted	Anchored	AgD	0.014	0.282	0.949	0.940	0.472
		Unanchored	AgD	0.121	0.237	0.932	0.882	0.271

Scenario number 2

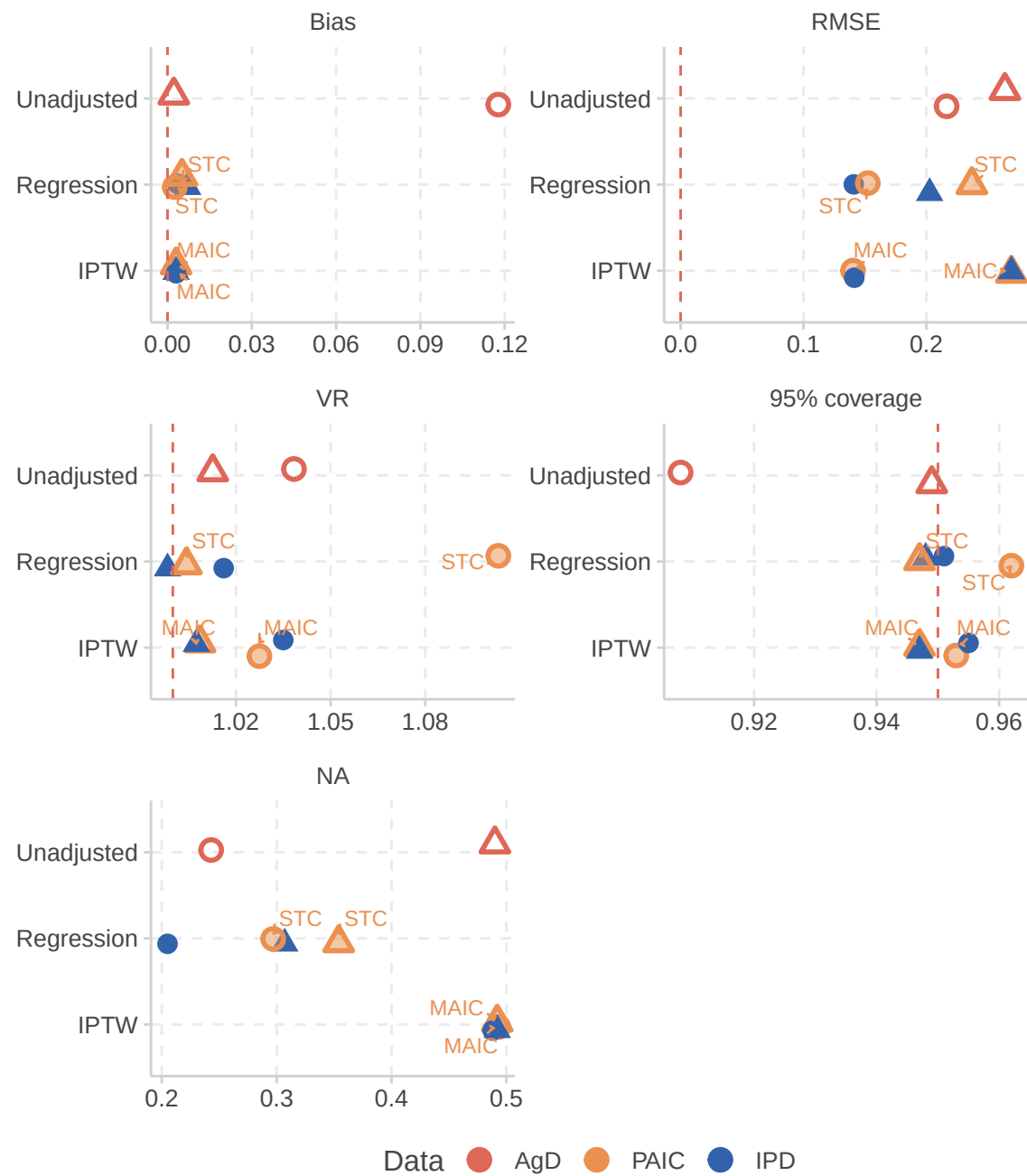
bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	0	0	$\text{rbinom}(N_{\text{pop}}, 1, 0.5)$	$\text{rnorm}(N_{\text{pop}}, 0.5, 1)$	X2	$Y_{\text{obs}} \sim X2 * \text{ttt}$



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	0.005	0.273	0.987	0.945	0.492
		Unanchored	PAIC	0.077	0.202	0.951	0.919	0.343
	ML	Anchored	IPD	0.005	0.273	0.988	0.945	0.491
		Unanchored	IPD	0.078	0.202	0.954	0.919	0.346
Regression								
	GLM	Anchored	IPD	0.003	0.253	0.989	0.935	0.364
		Unanchored	IPD	0.077	0.202	0.948	0.922	0.233
	STC	Anchored	PAIC	0.002	0.266	0.979	0.943	0.380
		Unanchored	PAIC	0.076	0.205	0.964	0.922	0.252
Unadjusted								
	Unadjusted	Anchored	AgD	0.004	0.274	0.977	0.938	0.493
		Unanchored	AgD	0.118	0.231	0.949	0.895	0.285

Scenario number 3

bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	0	0	rbinom(N_pop, 1, 0.5)	rnorm(N_pop, 0.5, 1)	X1, X2	Y_obs ~ X1*ttt + X2*ttt



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	0.003	0.269	1.009	0.947	0.492
		Unanchored	PAIC	0.003	0.140	1.027	0.953	0.493
	ML	Anchored	IPD	0.003	0.269	1.007	0.947	0.492
		Unanchored	IPD	0.003	0.141	1.035	0.955	0.488
Regression								
	GLM	Anchored	IPD	0.007	0.203	0.998	0.948	0.307
		Unanchored	IPD	0.003	0.141	1.016	0.951	0.205
	STC	Anchored	PAIC	0.005	0.237	1.004	0.947	0.354
		Unanchored	PAIC	0.003	0.152	1.103	0.962	0.297
Unadjusted								
	Unadjusted	Anchored	AgD	0.002	0.264	1.013	0.949	0.490
		Unanchored	AgD	0.118	0.217	1.038	0.908	0.243

Scenario number 4

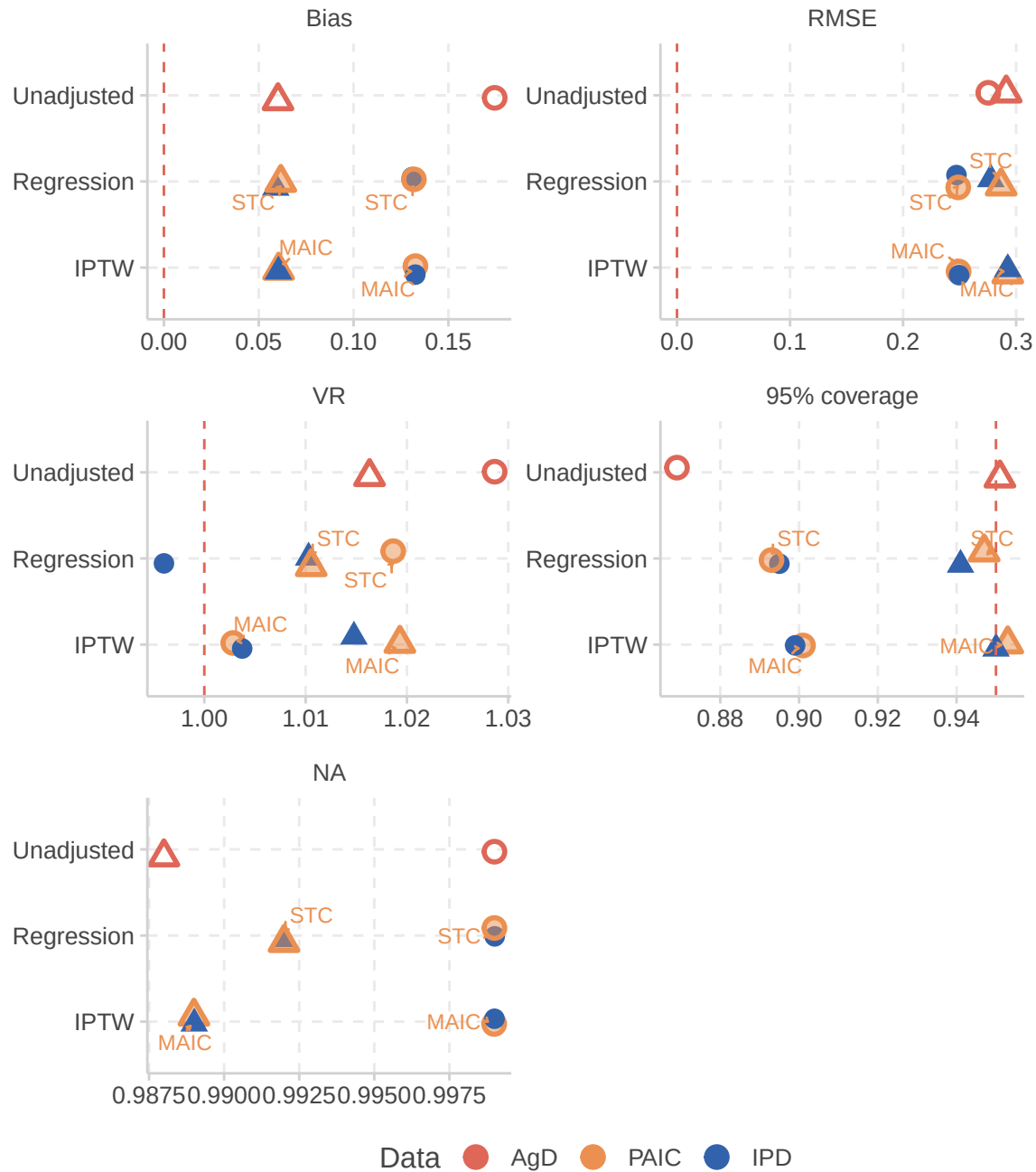
bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	1.2	0	$\text{rbinom}(N_{\text{pop}}, 1, 0.5)$	$\text{rnorm}(N_{\text{pop}}, 0.5, 1)$	X1	$Y_{\text{obs}} \sim X1 * \text{ttt}$



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	0.005	0.303	0.964	0.945	0.979
		Unanchored	PAIC	0.039	0.175	0.991	0.950	1.000
	ML	Anchored	IPD	0.005	0.303	0.963	0.941	0.979
		Unanchored	IPD	0.039	0.175	0.991	0.945	1.000
Regression								
	GLM	Anchored	IPD	-0.003	0.236	0.950	0.942	0.992
		Unanchored	IPD	0.037	0.170	0.954	0.942	1.000
	STC	Anchored	PAIC	-0.002	0.258	0.958	0.941	0.989
		Unanchored	PAIC	0.037	0.177	1.008	0.949	1.000
Unadjusted								
	Unadjusted	Anchored	AgD	0.068	0.309	0.961	0.936	0.984
		Unanchored	AgD	0.184	0.289	0.978	0.861	1.000

Scenario number 5

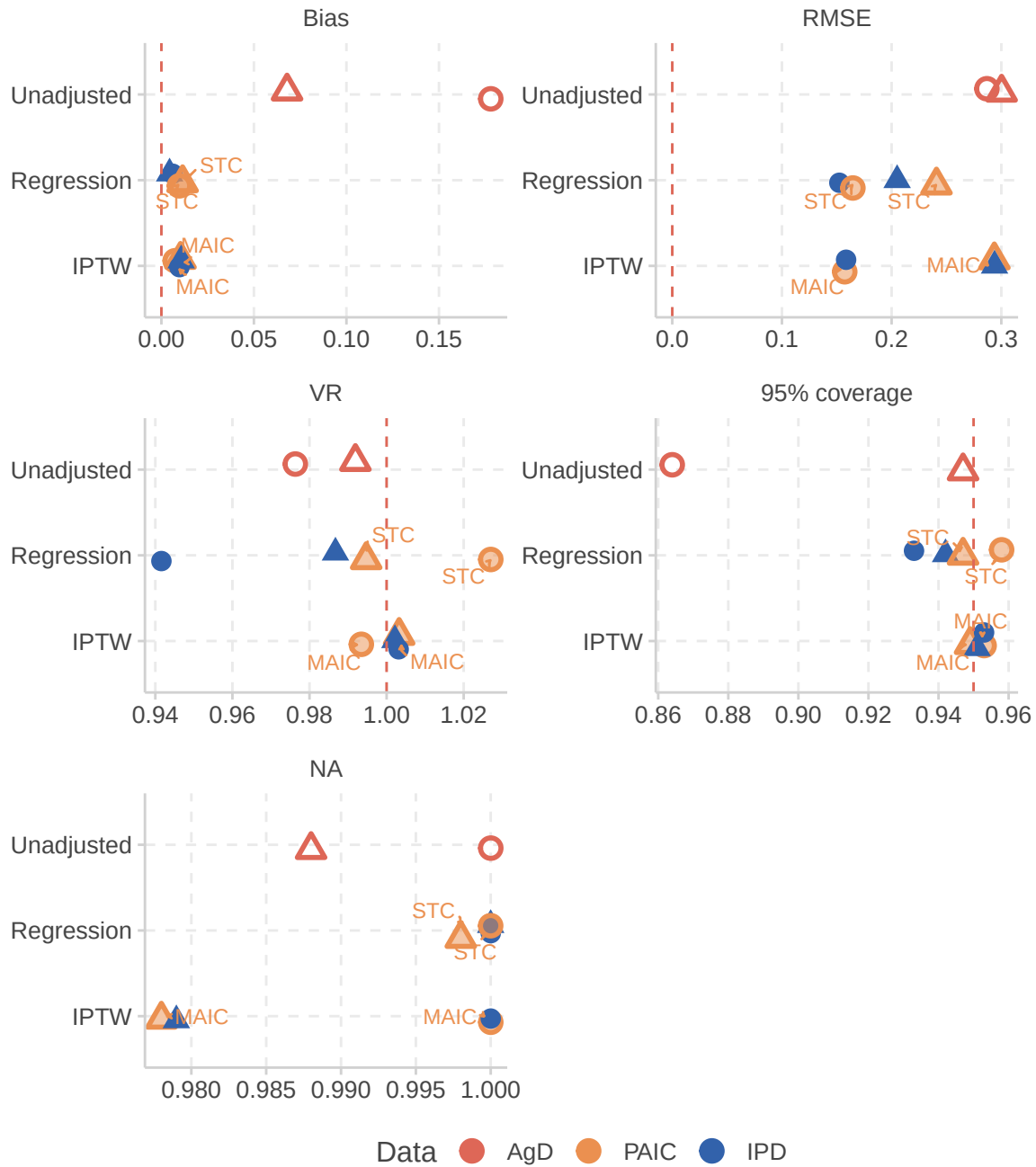
bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	1.2	0	$\text{rbinom}(N_{\text{pop}}, 1, 0.5)$	$\text{rnorm}(N_{\text{pop}}, 0.5, 1)$	X2	$Y_{\text{obs}} \sim X2 * \text{ttt}$



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	0.060	0.293	1.019	0.953	0.989
		Unanchored	PAIC	0.133	0.249	1.003	0.901	0.999
	ML	Anchored	IPD	0.060	0.293	1.015	0.950	0.989
		Unanchored	IPD	0.133	0.249	1.004	0.899	0.999
Regression								
	GLM	Anchored	IPD	0.059	0.278	1.010	0.941	0.992
		Unanchored	IPD	0.131	0.247	0.996	0.895	0.999
	STC	Anchored	PAIC	0.062	0.287	1.011	0.947	0.992
		Unanchored	PAIC	0.132	0.249	1.019	0.893	0.999
Unadjusted								
	Unadjusted	Anchored	AgD	0.060	0.291	1.016	0.951	0.988
		Unanchored	AgD	0.174	0.275	1.029	0.869	0.999

Scenario number 6

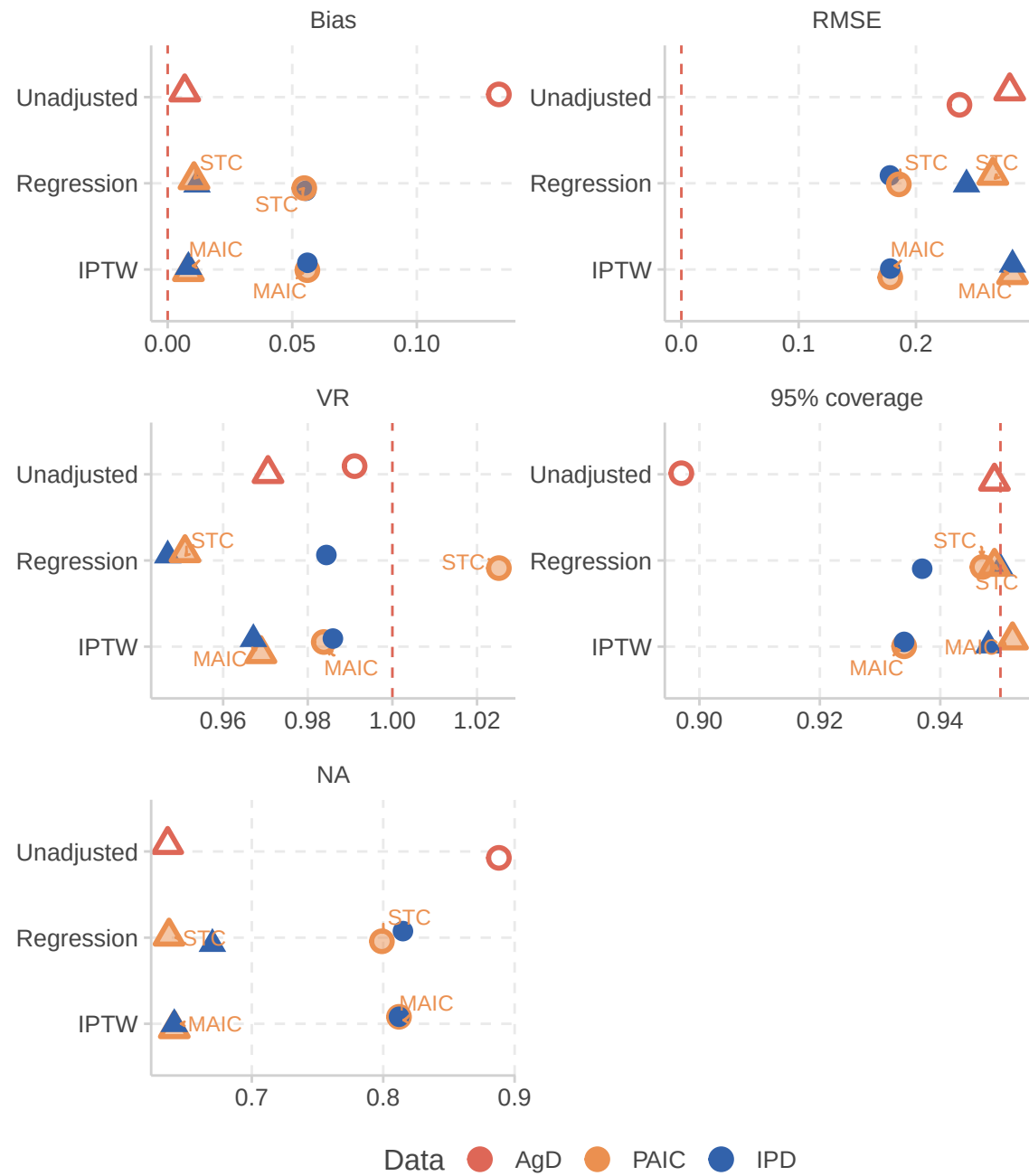
bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	1.2	0	$\text{rbinom}(N_{\text{pop}}, 1, 0.5)$	$\text{rnorm}(N_{\text{pop}}, 0.5, 1)$	X1, X2	$Y_{\text{obs}} \sim X1 * \text{ttt} + X2 * \text{ttt}$



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	0.010	0.294	1.003	0.949	0.978
		Unanchored	PAIC	0.007	0.157	0.993	0.953	1.000
	ML	Anchored	IPD	0.011	0.294	1.002	0.951	0.979
		Unanchored	IPD	0.010	0.159	1.003	0.953	1.000
Regression								
	GLM	Anchored	IPD	0.004	0.205	0.987	0.942	1.000
		Unanchored	IPD	0.006	0.152	0.942	0.933	1.000
	STC	Anchored	PAIC	0.011	0.241	0.995	0.947	0.998
		Unanchored	PAIC	0.010	0.165	1.027	0.958	1.000
Unadjusted								
	Unadjusted	Anchored	AgD	0.068	0.300	0.992	0.947	0.988
		Unanchored	AgD	0.178	0.287	0.976	0.864	1.000

Scenario number 7

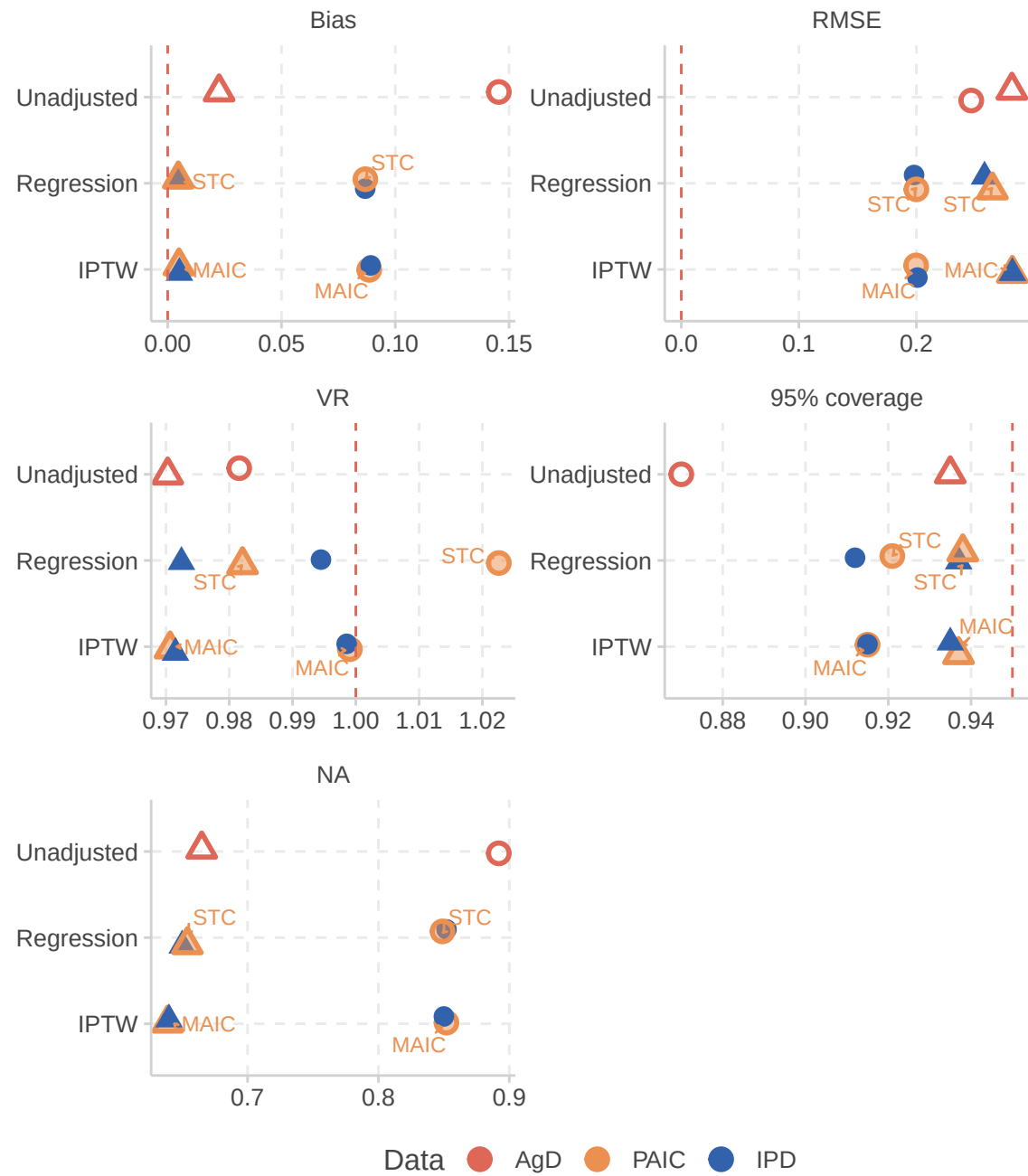
bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	0	0.2	rbinom(N_pop, 1, 0.5)	rnorm(N_pop, 0.5, 1)	X1	Y_obs ~ X1*ttt



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	0.008	0.282	0.969	0.952	0.641
		Unanchored	PAIC	0.056	0.178	0.984	0.934	0.812
	ML	Anchored	IPD	0.008	0.282	0.967	0.948	0.641
		Unanchored	IPD	0.056	0.178	0.986	0.934	0.812
Regression								
	GLM	Anchored	IPD	0.012	0.243	0.947	0.950	0.670
		Unanchored	IPD	0.056	0.178	0.984	0.937	0.815
	STC	Anchored	PAIC	0.011	0.266	0.951	0.949	0.637
		Unanchored	PAIC	0.055	0.185	1.025	0.947	0.799
Unadjusted								
	Unadjusted	Anchored	AgD	0.007	0.280	0.971	0.949	0.636
		Unanchored	AgD	0.133	0.237	0.991	0.897	0.888

Scenario number 8

bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	0	0.2	rbinom(N_pop, 1, 0.5)	rnorm(N_pop, 0.5, 1)	X2	Y_obs ~ X2*ttt



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	0.005	0.282	0.971	0.937	0.639
		Unanchored	PAIC	0.089	0.200	0.999	0.915	0.852
	ML	Anchored	IPD	0.005	0.282	0.971	0.935	0.640
		Unanchored	IPD	0.089	0.201	0.999	0.915	0.850
Regression								
	GLM	Anchored	IPD	0.004	0.258	0.972	0.937	0.650
		Unanchored	IPD	0.087	0.198	0.994	0.912	0.852
	STC	Anchored	PAIC	0.005	0.265	0.982	0.938	0.654
		Unanchored	PAIC	0.087	0.200	1.023	0.921	0.849
Unadjusted								
	Unadjusted	Anchored	AgD	0.023	0.281	0.970	0.935	0.665
		Unanchored	AgD	0.145	0.247	0.982	0.870	0.892

Scenario number 9

bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	0	0.2	rbinom(N_pop, 1, 0.5)	rnorm(N_pop, 0.5, 1)	X1, X2	$Y_{obs} \sim X1*ttt + X2*ttt$



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	0.002	0.286	0.964	0.941	0.633
		Unanchored	PAIC	-0.001	0.147	0.992	0.949	0.746
	ML	Anchored	IPD	0.002	0.286	0.963	0.935	0.634
		Unanchored	IPD	0.002	0.147	1.006	0.953	0.751
Regression								
	GLM	Anchored	IPD	-0.001	0.203	0.996	0.937	0.701
		Unanchored	IPD	-0.002	0.145	0.983	0.947	0.750
	STC	Anchored	PAIC	0.001	0.238	1.001	0.942	0.667
		Unanchored	PAIC	-0.001	0.158	1.070	0.961	0.722
Unadjusted								
	Unadjusted	Anchored	AgD	0.019	0.285	0.953	0.934	0.659
		Unanchored	AgD	0.129	0.240	0.964	0.898	0.869

Scenario number 10

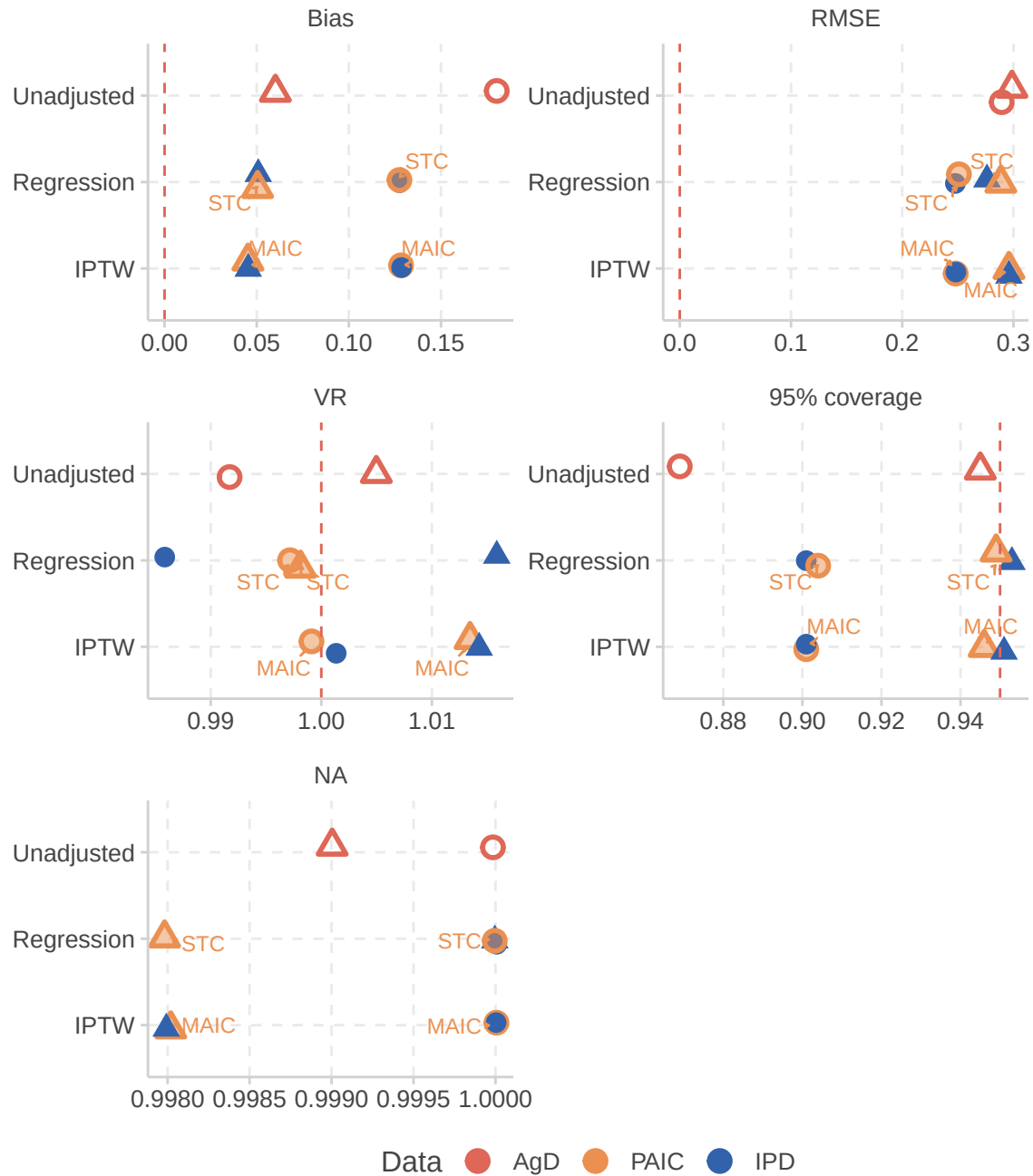
bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	1.2	0.2	$\text{rbinom}(N_{\text{pop}}, 1, 0.5)$	$\text{rnorm}(N_{\text{pop}}, 0.5, 1)$	X1	$Y_{\text{obs}} \sim X1 * \text{ttt}$



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	0.029	0.290	1.027	0.949	0.988
		Unanchored	PAIC	0.052	0.180	1.033	0.942	1.000
	ML	Anchored	IPD	0.029	0.290	1.027	0.947	0.988
		Unanchored	IPD	0.052	0.180	1.034	0.944	1.000
Regression								
	GLM	Anchored	IPD	0.018	0.228	1.011	0.949	0.999
		Unanchored	IPD	0.056	0.178	0.989	0.936	1.000
	STC	Anchored	PAIC	0.027	0.254	1.000	0.942	0.999
		Unanchored	PAIC	0.060	0.188	1.025	0.943	1.000
Unadjusted								
	Unadjusted	Anchored	AgD	0.088	0.304	1.011	0.943	0.992
		Unanchored	AgD	0.193	0.298	0.992	0.855	1.000

Scenario number 11

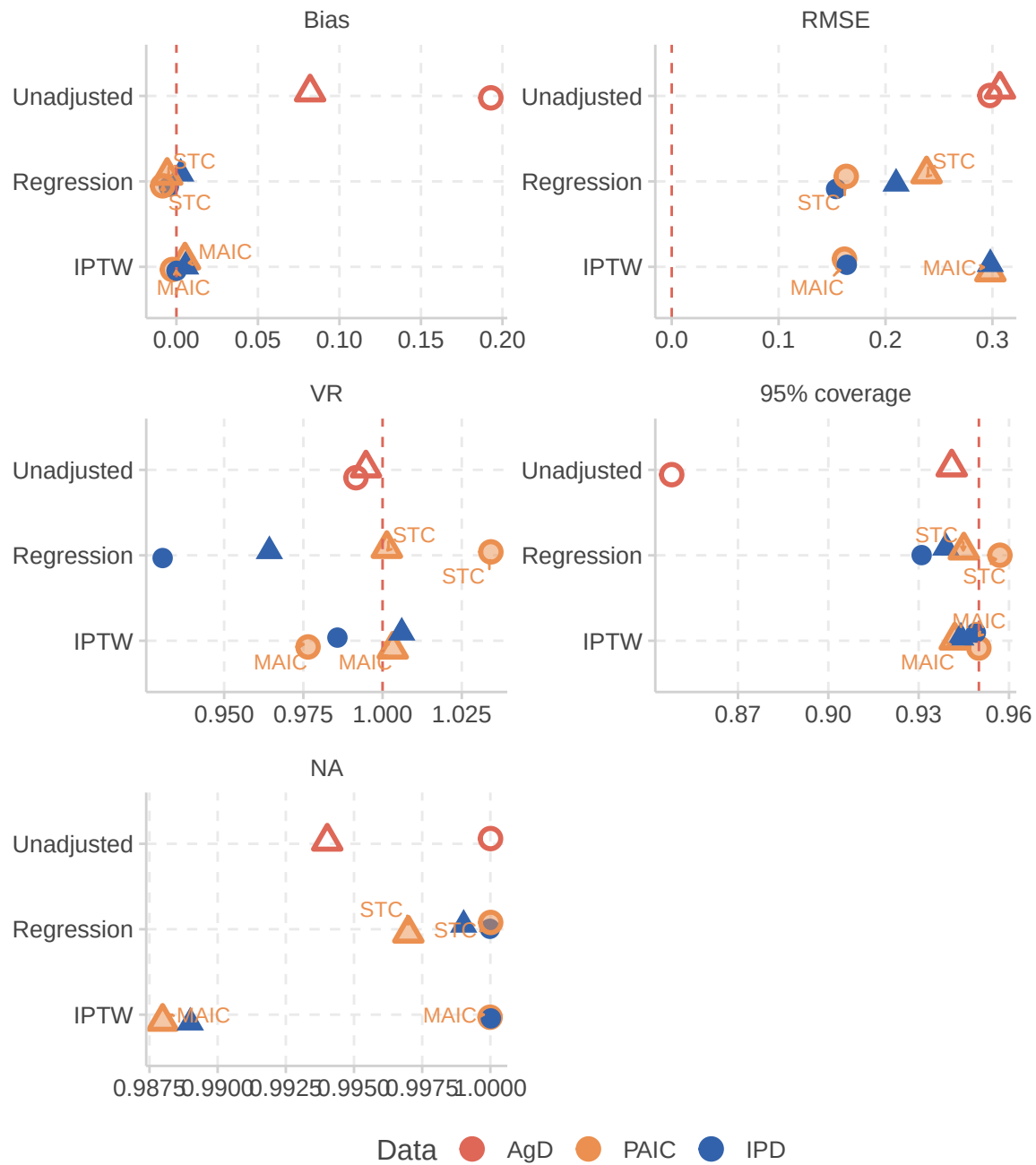
bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	1.2	0.2	$\text{rbinom}(N_{\text{pop}}, 1, 0.5)$	$\text{rnorm}(N_{\text{pop}}, 0.5, 1)$	X2	$Y_{\text{obs}} \sim X2 * \text{ttt}$



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	0.045	0.296	1.013	0.946	0.998
		Unanchored	PAIC	0.128	0.248	0.999	0.901	1.000
	ML	Anchored	IPD	0.045	0.296	1.014	0.951	0.998
		Unanchored	IPD	0.129	0.248	1.001	0.901	1.000
Regression								
	GLM	Anchored	IPD	0.051	0.276	1.016	0.953	1.000
		Unanchored	IPD	0.128	0.248	0.986	0.901	1.000
	STC	Anchored	PAIC	0.050	0.289	0.998	0.949	0.998
		Unanchored	PAIC	0.127	0.251	0.997	0.904	1.000
Unadjusted								
	Unadjusted	Anchored	AgD	0.060	0.299	1.005	0.945	0.999
		Unanchored	AgD	0.180	0.289	0.992	0.869	1.000

Scenario number 12

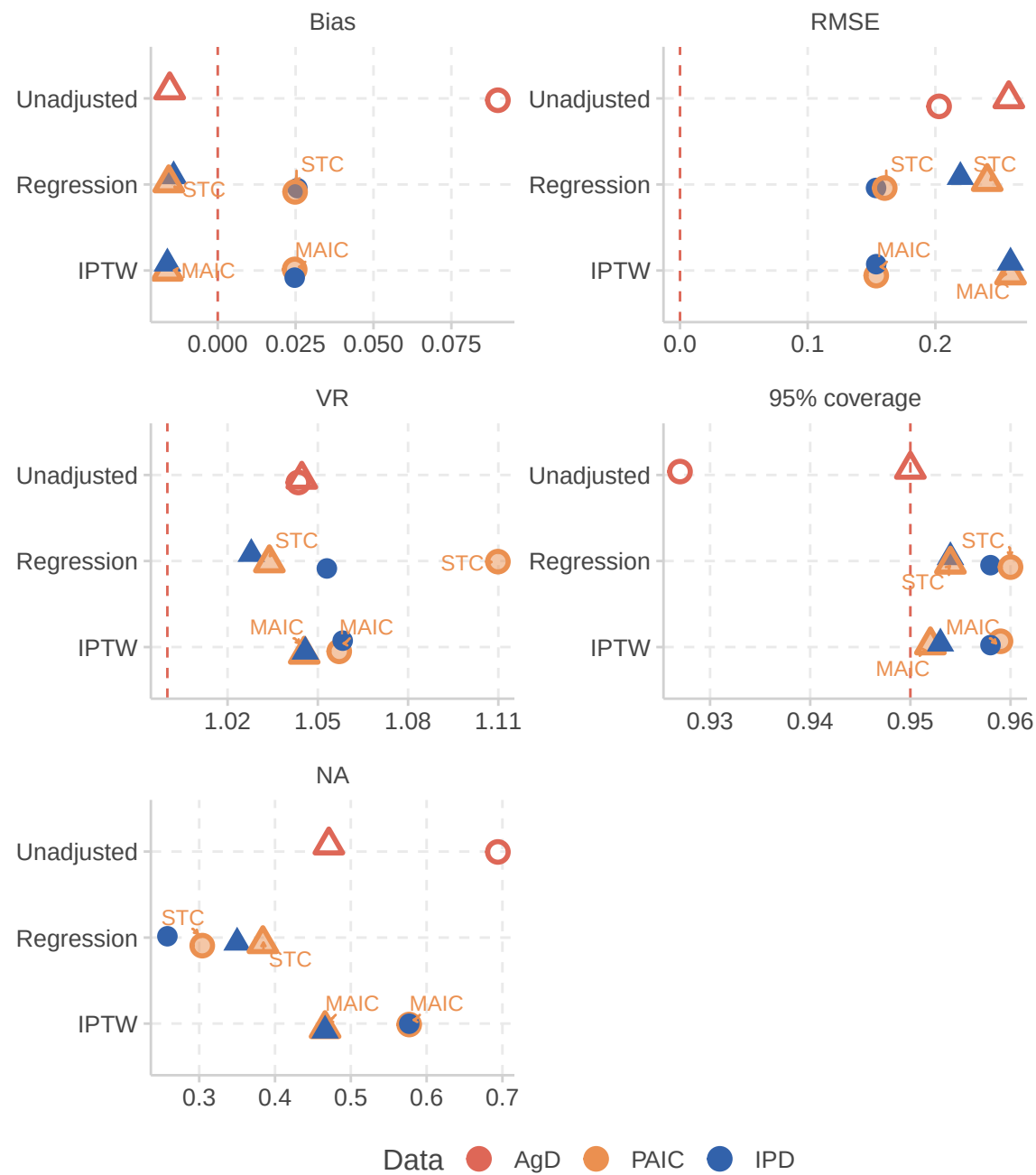
bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	1.2	0.2	$\text{rbinom}(N_{\text{pop}}, 1, 0.5)$	$\text{rnorm}(N_{\text{pop}}, 0.5, 1)$	X1, X2	$Y_{\text{obs}} \sim X1 * \text{ttt} + X2 * \text{ttt}$



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	0.005	0.298	1.003	0.942	0.988
		Unanchored	PAIC	-0.002	0.162	0.976	0.950	1.000
	ML	Anchored	IPD	0.006	0.298	1.006	0.944	0.989
		Unanchored	IPD	0.000	0.164	0.986	0.949	1.000
Regression								
	GLM	Anchored	IPD	0.003	0.210	0.964	0.939	0.999
		Unanchored	IPD	-0.005	0.154	0.931	0.931	1.000
	STC	Anchored	PAIC	-0.006	0.238	1.001	0.945	0.997
		Unanchored	PAIC	-0.008	0.163	1.034	0.957	1.000
Unadjusted								
	Unadjusted	Anchored	AgD	0.082	0.307	0.995	0.941	0.994
		Unanchored	AgD	0.193	0.298	0.992	0.848	1.000

Scenario number 13

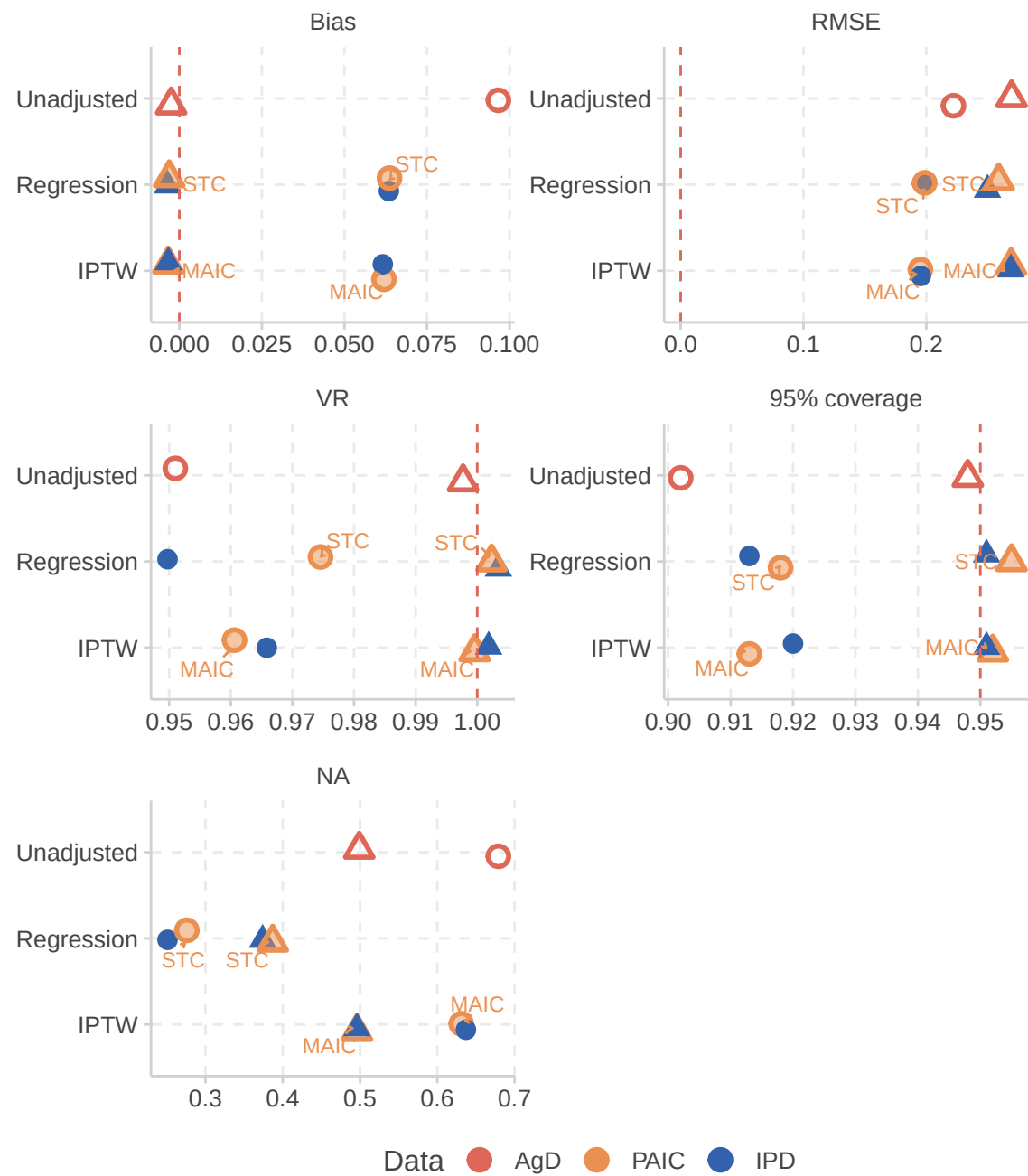
bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	0	0	rbinom(N_pop, 1, 0.5)	rlnorm(N_pop, 0.5, 0.5)	X1	Y_obs ~ X1*ttt



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	-0.016	0.259	1.045	0.952	0.466
		Unanchored	PAIC	0.025	0.154	1.057	0.959	0.577
	ML	Anchored	IPD	-0.016	0.259	1.046	0.953	0.466
		Unanchored	IPD	0.025	0.154	1.058	0.958	0.577
Regression								
	GLM	Anchored	IPD	-0.014	0.220	1.028	0.954	0.350
		Unanchored	IPD	0.026	0.154	1.053	0.958	0.258
	STC	Anchored	PAIC	-0.016	0.241	1.034	0.954	0.384
		Unanchored	PAIC	0.025	0.160	1.110	0.960	0.304
Unadjusted								
	Unadjusted	Anchored	AgD	-0.015	0.258	1.045	0.950	0.471
		Unanchored	AgD	0.090	0.203	1.044	0.927	0.694

Scenario number 14

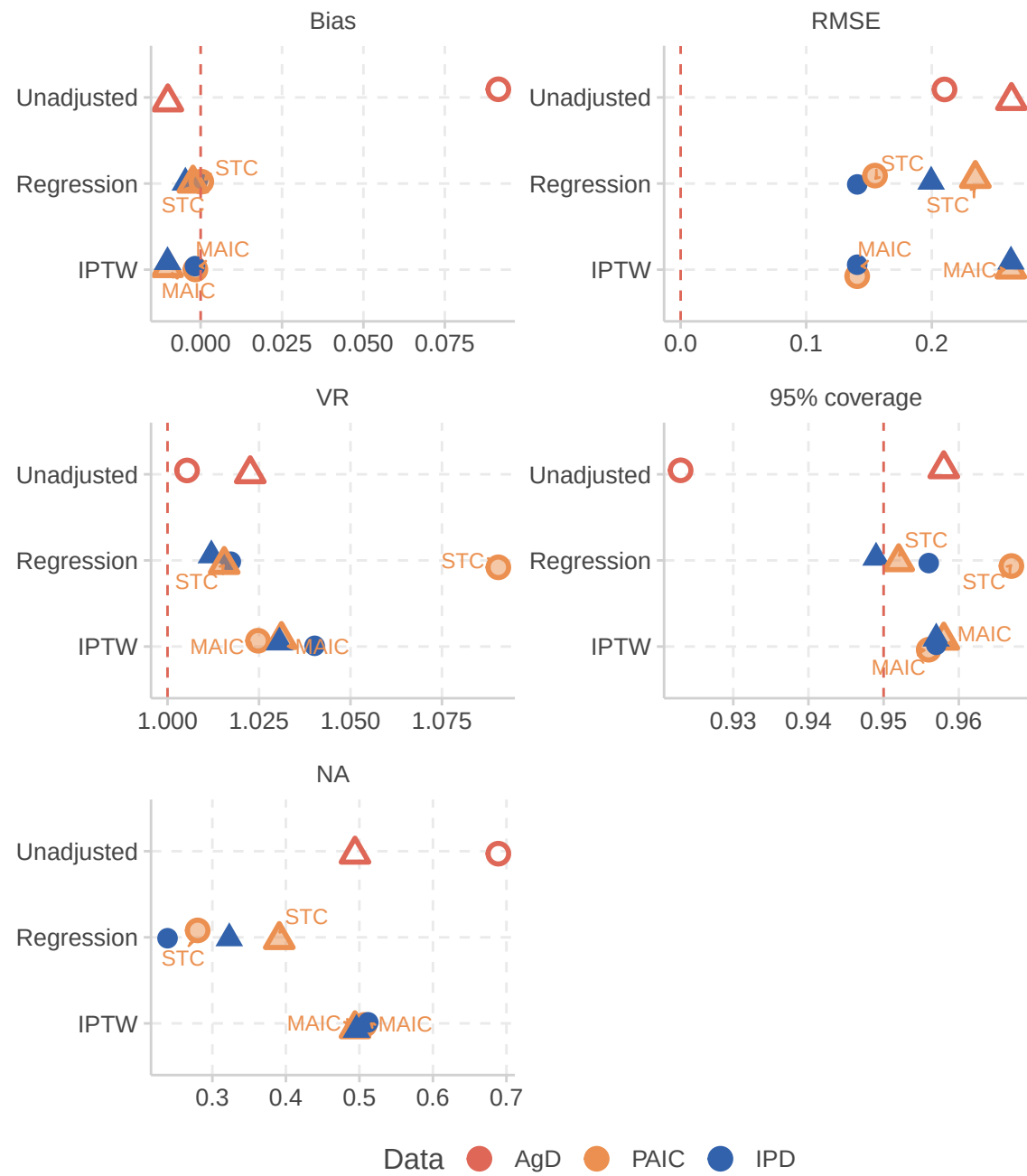
bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	0	0	rbinom(N_pop, 1, 0.5)	rlnorm(N_pop, 0.5, 0.5)	X2	Y_obs ~ X2*ttt



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	-0.003	0.269	1.000	0.952	0.496
		Unanchored	PAIC	0.062	0.195	0.961	0.913	0.631
	ML	Anchored	IPD	-0.003	0.269	1.002	0.951	0.496
		Unanchored	IPD	0.062	0.196	0.966	0.920	0.637
Regression								
	GLM	Anchored	IPD	-0.004	0.250	1.003	0.951	0.374
		Unanchored	IPD	0.063	0.197	0.950	0.913	0.251
	STC	Anchored	PAIC	-0.003	0.259	1.002	0.955	0.387
		Unanchored	PAIC	0.064	0.199	0.975	0.918	0.276
Unadjusted								
	Unadjusted	Anchored	AgD	-0.002	0.269	0.998	0.948	0.499
		Unanchored	AgD	0.097	0.222	0.951	0.902	0.679

Scenario number 15

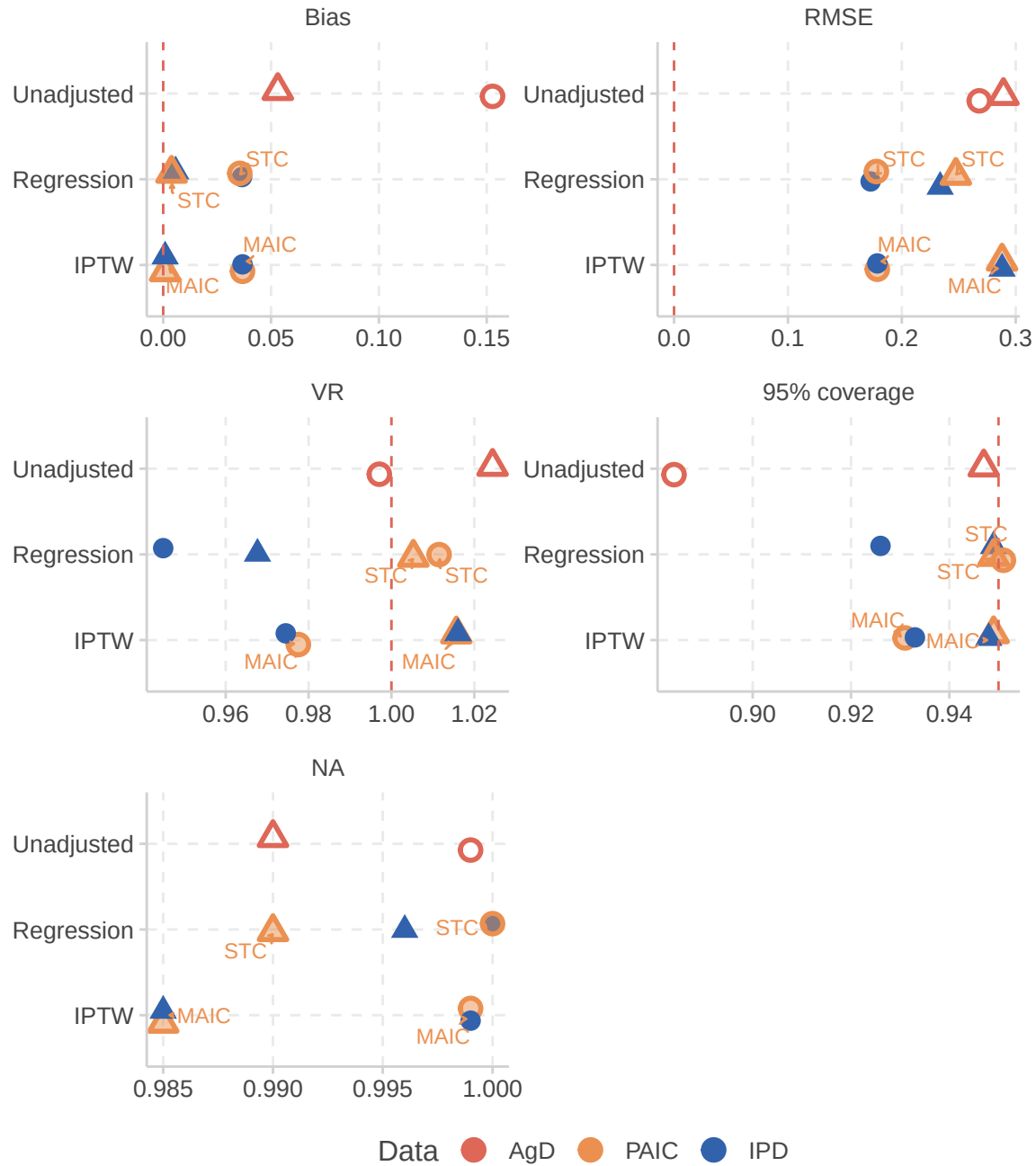
bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	0	0	rbinom(N_pop, 1, 0.5)	rlnorm(N_pop, 0.5, 0.5)	X1, X2	Y_obs ~ X1*ttt + X2*ttt



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	-0.010	0.263	1.031	0.958	0.494
		Unanchored	PAIC	-0.002	0.141	1.025	0.956	0.507
	ML	Anchored	IPD	-0.010	0.263	1.031	0.957	0.496
		Unanchored	IPD	-0.002	0.141	1.040	0.957	0.512
Regression								
	GLM	Anchored	IPD	-0.005	0.199	1.012	0.949	0.323
		Unanchored	IPD	-0.001	0.141	1.017	0.956	0.239
	STC	Anchored	PAIC	-0.002	0.235	1.015	0.952	0.391
		Unanchored	PAIC	0.000	0.155	1.090	0.967	0.280
Unadjusted								
	Unadjusted	Anchored	AgD	-0.010	0.263	1.023	0.958	0.494
		Unanchored	AgD	0.091	0.210	1.005	0.923	0.689

Scenario number 16

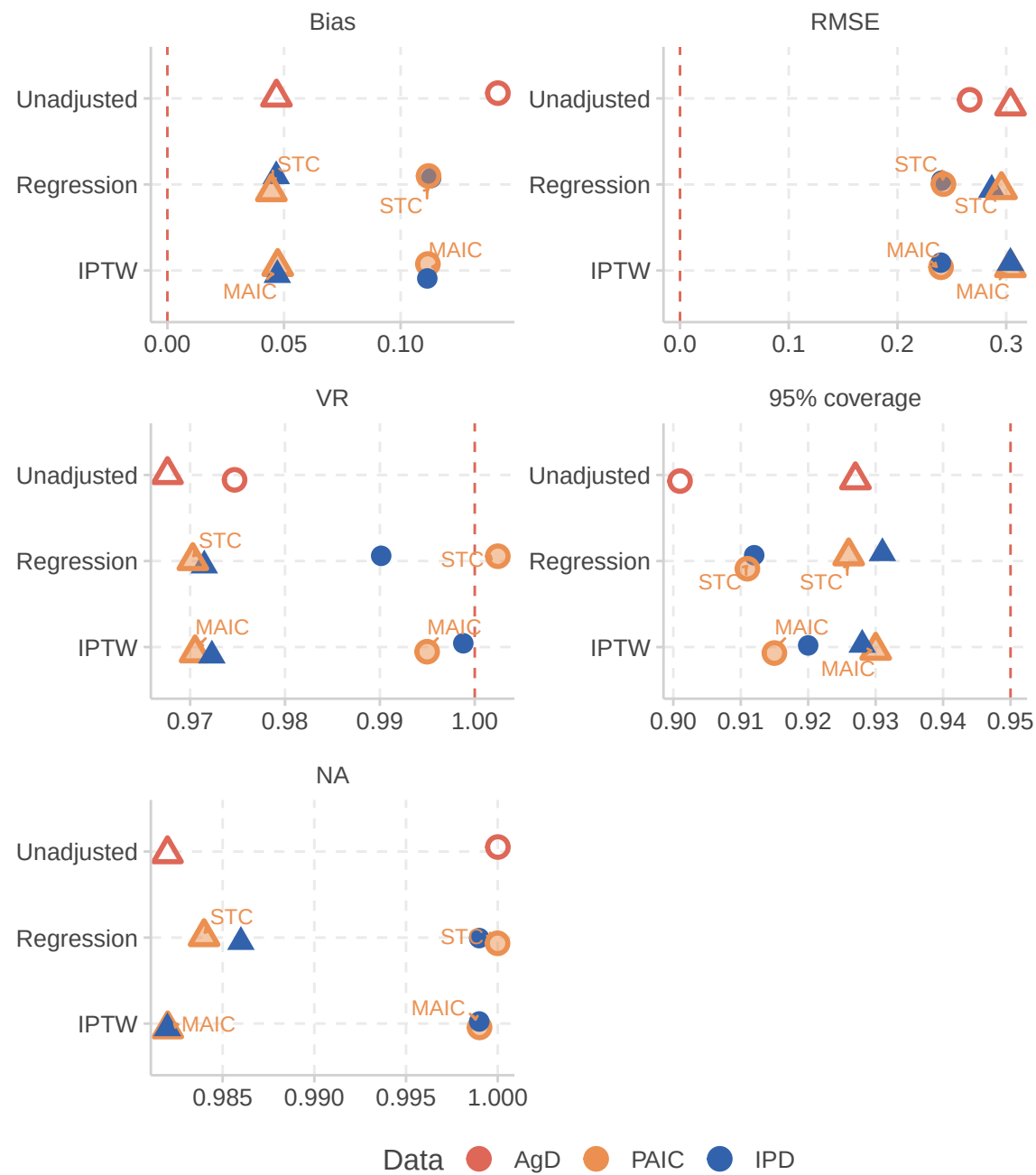
bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	1.2	0	$\text{rbinom}(N_{\text{pop}}, 1, 0.5)$	$\text{rlnorm}(N_{\text{pop}}, 0.5, 0.5)$	X1	$Y_{\text{obs}} \sim X1 * \text{ttt}$



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	0.001	0.288	1.016	0.949	0.985
		Unanchored	PAIC	0.037	0.179	0.978	0.931	0.999
	ML	Anchored	IPD	0.001	0.288	1.016	0.948	0.985
		Unanchored	IPD	0.037	0.179	0.974	0.933	0.999
Regression								
	GLM	Anchored	IPD	0.006	0.234	0.968	0.949	0.996
		Unanchored	IPD	0.037	0.173	0.945	0.926	1.000
	STC	Anchored	PAIC	0.004	0.247	1.005	0.949	0.990
		Unanchored	PAIC	0.036	0.178	1.012	0.951	1.000
Unadjusted								
	Unadjusted	Anchored	AgD	0.053	0.289	1.024	0.947	0.990
		Unanchored	AgD	0.153	0.268	0.997	0.884	0.999

Scenario number 17

bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	1.2	0	rbinom(N_pop, 1, 0.5)	rlnorm(N_pop, 0.5, 0.5)	X2	Y_obs ~ X2*ttt



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	0.047	0.304	0.971	0.930	0.982
		Unanchored	PAIC	0.112	0.240	0.995	0.915	0.999
	ML	Anchored	IPD	0.047	0.304	0.972	0.928	0.982
		Unanchored	IPD	0.111	0.240	0.999	0.920	0.999
Regression								
	GLM	Anchored	IPD	0.047	0.287	0.971	0.931	0.986
		Unanchored	IPD	0.113	0.240	0.990	0.912	0.999
	STC	Anchored	PAIC	0.045	0.296	0.970	0.926	0.984
		Unanchored	PAIC	0.112	0.242	1.002	0.911	1.000
Unadjusted								
	Unadjusted	Anchored	AgD	0.047	0.304	0.968	0.927	0.982
		Unanchored	AgD	0.142	0.266	0.975	0.901	1.000

Scenario number 18

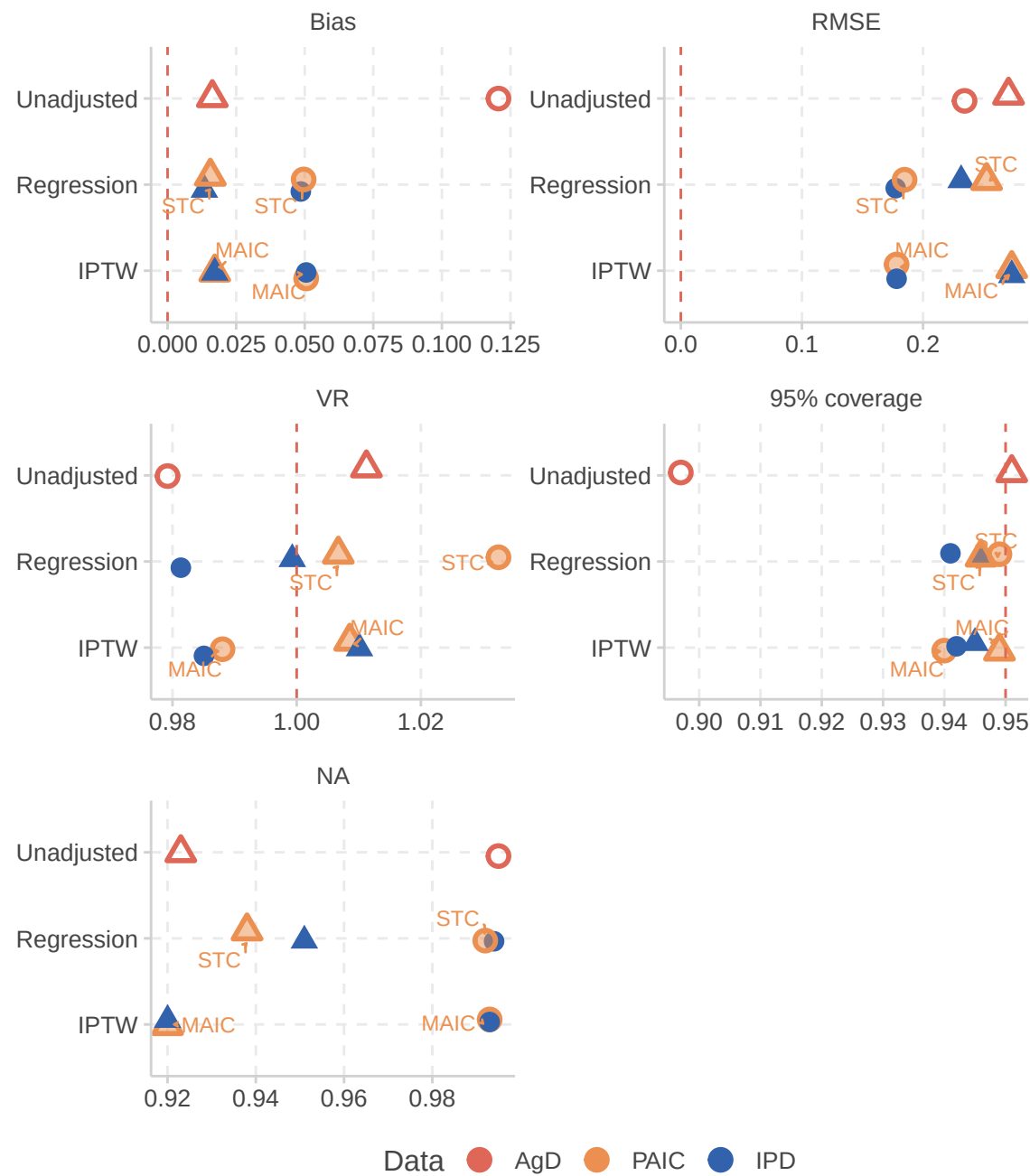
bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	1.2	0	rbinom(N_pop, 1, 0.5)	rlnorm(N_pop, 0.5, 0.5)	X1, X2	Y_obs ~ X1*ttt + X2*ttt



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	0.011	0.297	0.988	0.942	0.981
		Unanchored	PAIC	0.001	0.160	0.974	0.937	1.000
	ML	Anchored	IPD	0.011	0.297	0.991	0.943	0.982
		Unanchored	IPD	0.001	0.159	0.992	0.948	1.000
Regression								
	GLM	Anchored	IPD	0.004	0.213	0.947	0.939	0.998
		Unanchored	IPD	0.003	0.154	0.928	0.930	1.000
	STC	Anchored	PAIC	0.010	0.246	0.969	0.947	0.995
		Unanchored	PAIC	0.006	0.165	1.019	0.954	1.000
Unadjusted								
	Unadjusted	Anchored	AgD	0.062	0.306	0.970	0.936	0.987
		Unanchored	AgD	0.149	0.273	0.959	0.903	1.000

Scenario number 19

bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	0	0.2	rbinom(N_pop, 1, 0.5)	rlnorm(N_pop, 0.5, 0.5)	X1	Y_obs ~ X1*ttt



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	0.017	0.273	1.009	0.949	0.920
		Unanchored	PAIC	0.051	0.178	0.988	0.940	0.993
	ML	Anchored	IPD	0.017	0.273	1.010	0.945	0.920
		Unanchored	IPD	0.051	0.178	0.985	0.942	0.993
Regression								
	GLM	Anchored	IPD	0.013	0.231	0.999	0.946	0.951
		Unanchored	IPD	0.049	0.177	0.981	0.941	0.994
	STC	Anchored	PAIC	0.016	0.252	1.007	0.946	0.938
		Unanchored	PAIC	0.050	0.185	1.032	0.949	0.992
Unadjusted								
	Unadjusted	Anchored	AgD	0.016	0.271	1.011	0.951	0.923
		Unanchored	AgD	0.121	0.234	0.979	0.897	0.995

Scenario number 20

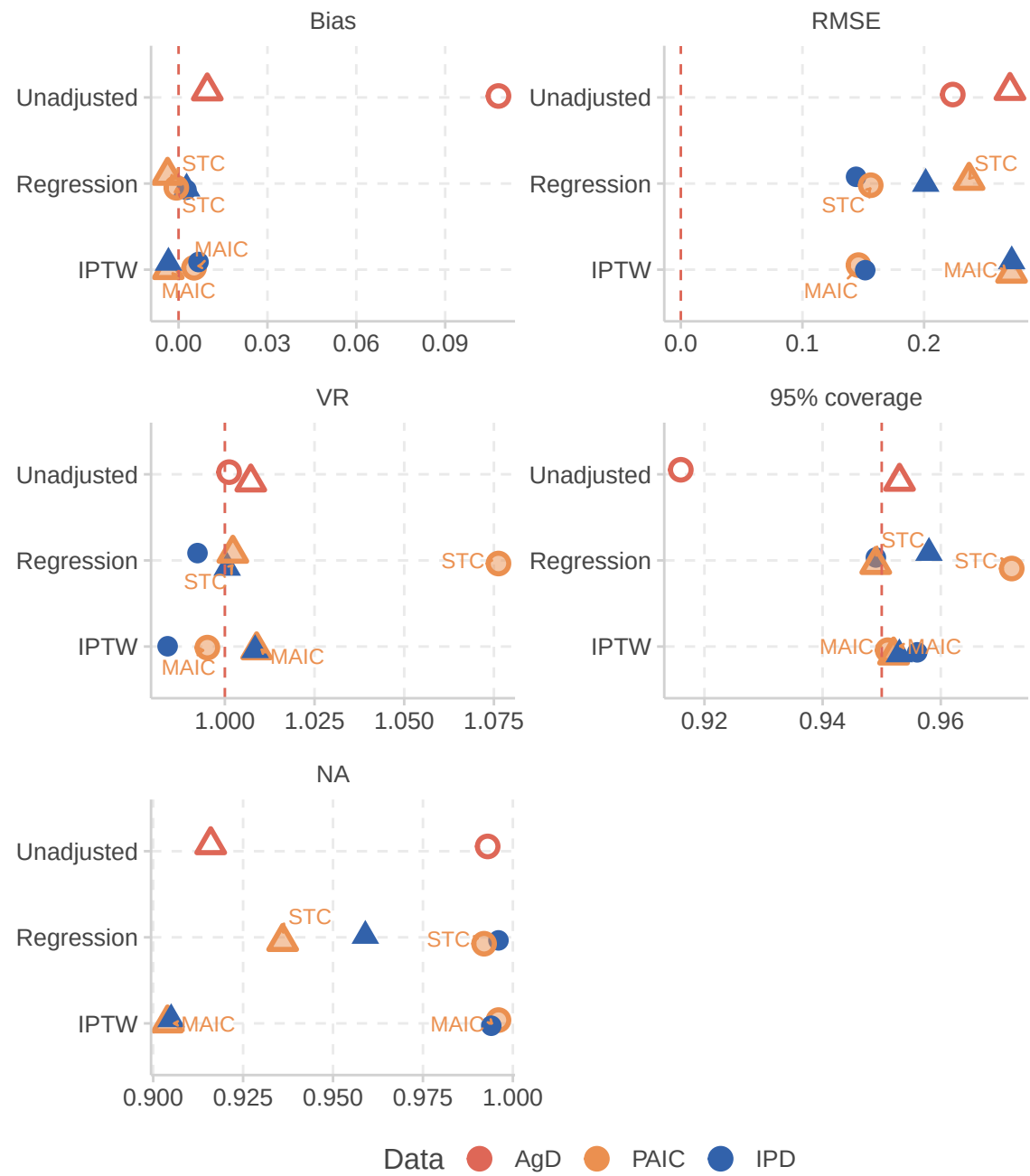
bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	0	0.2	$\text{rbinom}(N_{\text{pop}}, 1, 0.5)$	$\text{rlnorm}(N_{\text{pop}}, 0.5, 0.5)$	X2	$Y_{\text{obs}} \sim X2 * \text{ttt}$



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	-0.005	0.271	1.008	0.960	0.915
		Unanchored	PAIC	0.059	0.192	0.978	0.923	0.988
	ML	Anchored	IPD	-0.005	0.271	1.010	0.962	0.915
		Unanchored	IPD	0.058	0.192	0.990	0.934	0.989
Regression								
	GLM	Anchored	IPD	-0.011	0.253	0.991	0.953	0.926
		Unanchored	IPD	0.056	0.191	0.974	0.931	0.988
	STC	Anchored	PAIC	-0.010	0.257	1.015	0.962	0.922
		Unanchored	PAIC	0.057	0.193	0.995	0.944	0.988
Unadjusted								
	Unadjusted	Anchored	AgD	0.006	0.272	1.003	0.955	0.923
		Unanchored	AgD	0.101	0.224	0.982	0.913	0.993

Scenario number 21

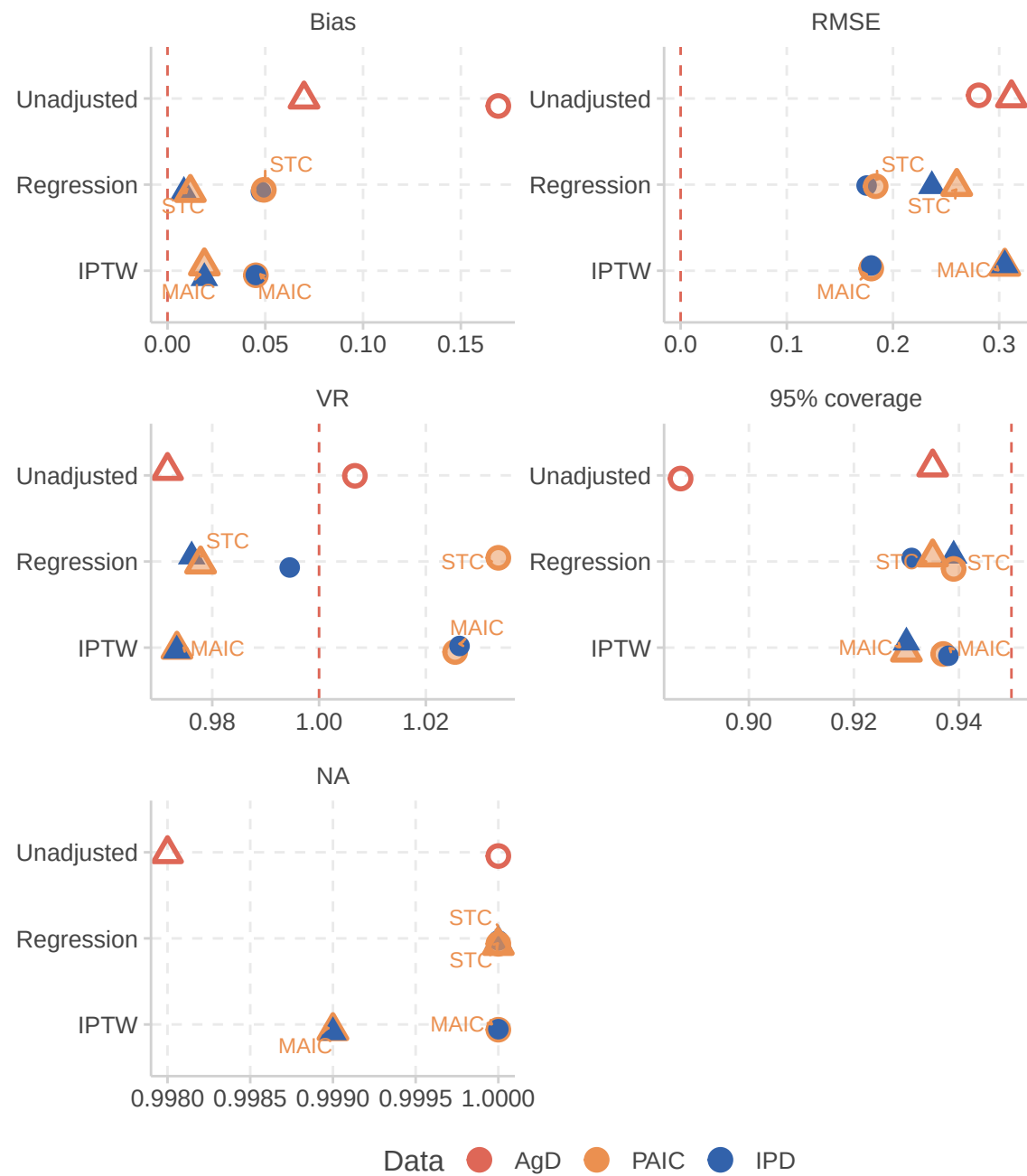
bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	0	0.2	rbinom(N_pop, 1, 0.5)	rlnorm(N_pop, 0.5, 0.5)	X1, X2	$Y_{obs} \sim X1*ttt + X2*ttt$



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	-0.003	0.272	1.009	0.952	0.904
		Unanchored	PAIC	0.005	0.146	0.995	0.951	0.996
	ML	Anchored	IPD	-0.003	0.272	1.008	0.953	0.905
		Unanchored	IPD	0.007	0.152	0.984	0.956	0.994
Regression								
	GLM	Anchored	IPD	0.003	0.201	1.001	0.958	0.959
		Unanchored	IPD	0.003	0.144	0.992	0.949	0.996
	STC	Anchored	PAIC	-0.004	0.237	1.002	0.949	0.936
		Unanchored	PAIC	-0.001	0.156	1.076	0.972	0.992
Unadjusted								
	Unadjusted	Anchored	AgD	0.010	0.271	1.007	0.953	0.916
		Unanchored	AgD	0.108	0.224	1.001	0.916	0.993

Scenario number 22

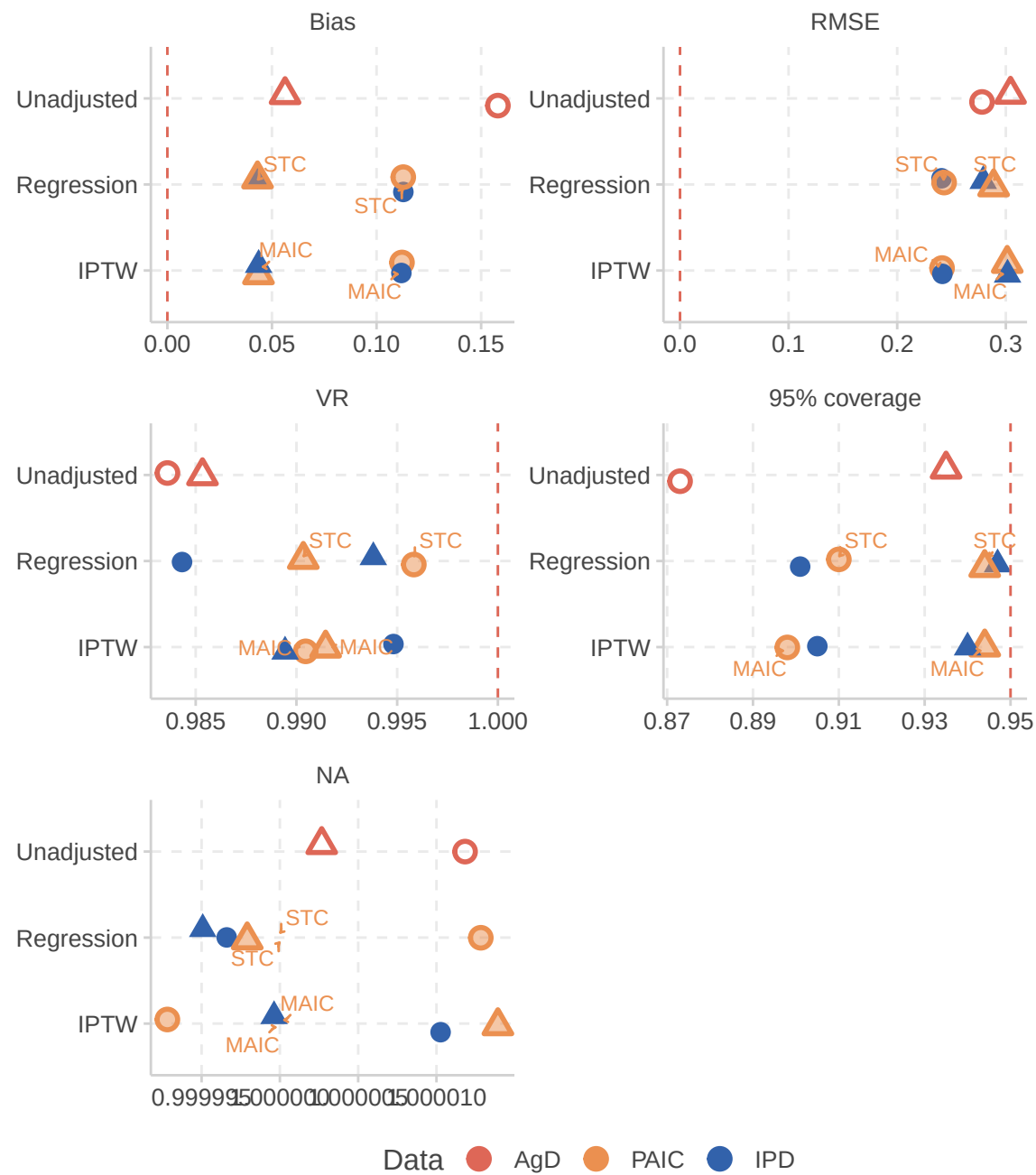
bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	1.2	0.2	rbinom(N_pop, 1, 0.5)	rlnorm(N_pop, 0.5, 0.5)	X1	Y_obs ~ X1*ttt



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	0.019	0.305	0.973	0.930	0.999
		Unanchored	PAIC	0.045	0.180	1.025	0.937	1.000
	ML	Anchored	IPD	0.019	0.305	0.973	0.930	0.999
		Unanchored	IPD	0.045	0.180	1.026	0.938	1.000
Regression								
	GLM	Anchored	IPD	0.008	0.237	0.976	0.939	1.000
		Unanchored	IPD	0.048	0.175	0.995	0.931	1.000
	STC	Anchored	PAIC	0.012	0.260	0.978	0.935	1.000
		Unanchored	PAIC	0.049	0.184	1.034	0.939	1.000
Unadjusted								
	Unadjusted	Anchored	AgD	0.070	0.312	0.972	0.935	0.998
		Unanchored	AgD	0.169	0.281	1.007	0.887	1.000

Scenario number 23

bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	1.2	0.2	rbinom(N_pop, 1, 0.5)	rlnorm(N_pop, 0.5, 0.5)	X2	Y_obs ~ X2*ttt



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	0.044	0.301	0.991	0.944	1.000
		Unanchored	PAIC	0.112	0.241	0.990	0.898	1.000
	ML	Anchored	IPD	0.044	0.302	0.989	0.940	1.000
		Unanchored	IPD	0.112	0.242	0.995	0.905	1.000
Regression								
	GLM	Anchored	IPD	0.042	0.279	0.994	0.947	1.000
		Unanchored	IPD	0.113	0.241	0.984	0.901	1.000
	STC	Anchored	PAIC	0.043	0.289	0.990	0.944	1.000
		Unanchored	PAIC	0.113	0.243	0.996	0.910	1.000
Unadjusted								
	Unadjusted	Anchored	AgD	0.056	0.304	0.985	0.935	1.000
		Unanchored	AgD	0.158	0.278	0.984	0.873	1.000

Scenario number 24

bY_X1	bY_X2	bY_A_X1	bY_A_X2	f_X1	f_X2	PS_model	regression
1.5	0.5	1.2	0.2	rbinom(N_pop, 1, 0.5)	rlnorm(N_pop, 0.5, 0.5)	X1, X2	Y_obs ~ X1*ttt + X2*ttt



	Model	Anchored	Data	Bias	RMSE	VR	95% coverage	NA
IPTW								
	MAIC	Anchored	PAIC	0.024	0.300	0.998	0.943	1.000
		Unanchored	PAIC	0.004	0.153	1.024	0.948	1.000
	ML	Anchored	IPD	0.024	0.300	0.998	0.939	1.000
		Unanchored	IPD	0.004	0.154	1.046	0.953	1.000
Regression								
	GLM	Anchored	IPD	0.014	0.203	0.997	0.939	1.000
		Unanchored	IPD	0.006	0.149	0.957	0.931	1.000
	STC	Anchored	PAIC	0.017	0.249	0.957	0.934	1.000
		Unanchored	PAIC	0.008	0.165	1.022	0.959	1.000
Unadjusted								
	Unadjusted	Anchored	AgD	0.090	0.312	0.988	0.930	1.000
		Unanchored	AgD	0.179	0.285	1.019	0.878	1.000